

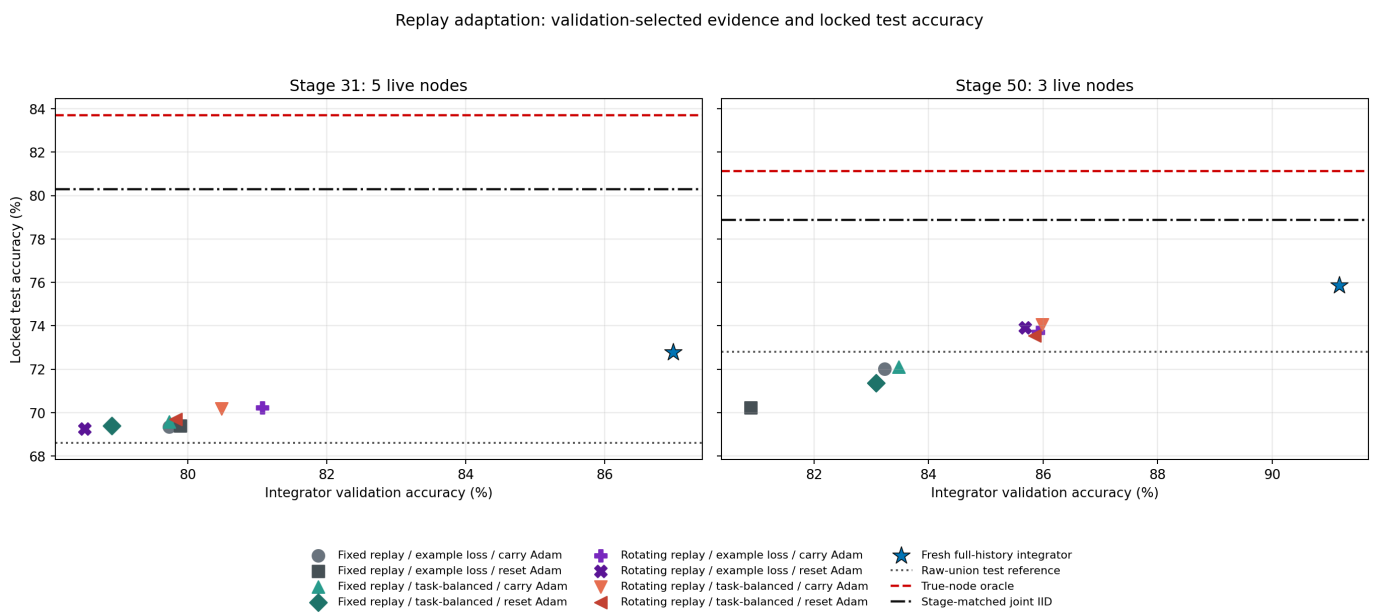
ImageNet-R-50 replay-adaptation diagnosis

A paired test of stage-random replay, equal task weighting, optimizer-state resets, and fresh full-history integration at fragmented frontiers.

Finding. Permuted-MNIST used a fresh deterministic replay draw at every macro-step. The validation-selected online ImageNet-R condition is **Rotating replay / example loss / carry Adam**. Averaged across matched cells and stages 31/50, rotation changed locked-test accuracy by +1.395 percentage points.

Headline comparison

Stage	Fixed baseline	Best online	Fresh full history	Raw union	True-node oracle	Joint IID
31	69.340	70.231	72.773	68.606	83.700	80.294
50	72.017	73.700	75.867	72.800	81.117	78.867

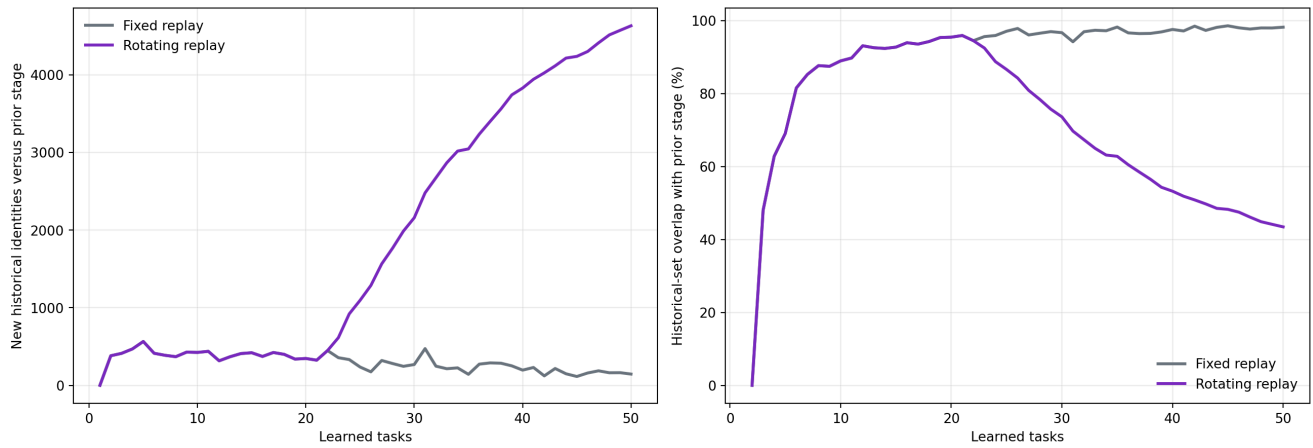


All online conditions and the validation-selected full-history diagnostic. Horizontal references use the identical stage and test examples.

What differed from Permuted-MNIST

Permuted-MNIST sampled uniformly from its complete historical archive with a seed containing the macro-step. It used 256 current and 256 historical examples. ImageNet-R instead used a permanent-priority namespace independent of stage. Stage-keyed random replay remains deterministic and resumable while preserving fixed-H cumulative $O(T \log T)$ work.

Stage-keyed replay changes which old examples train the integrator



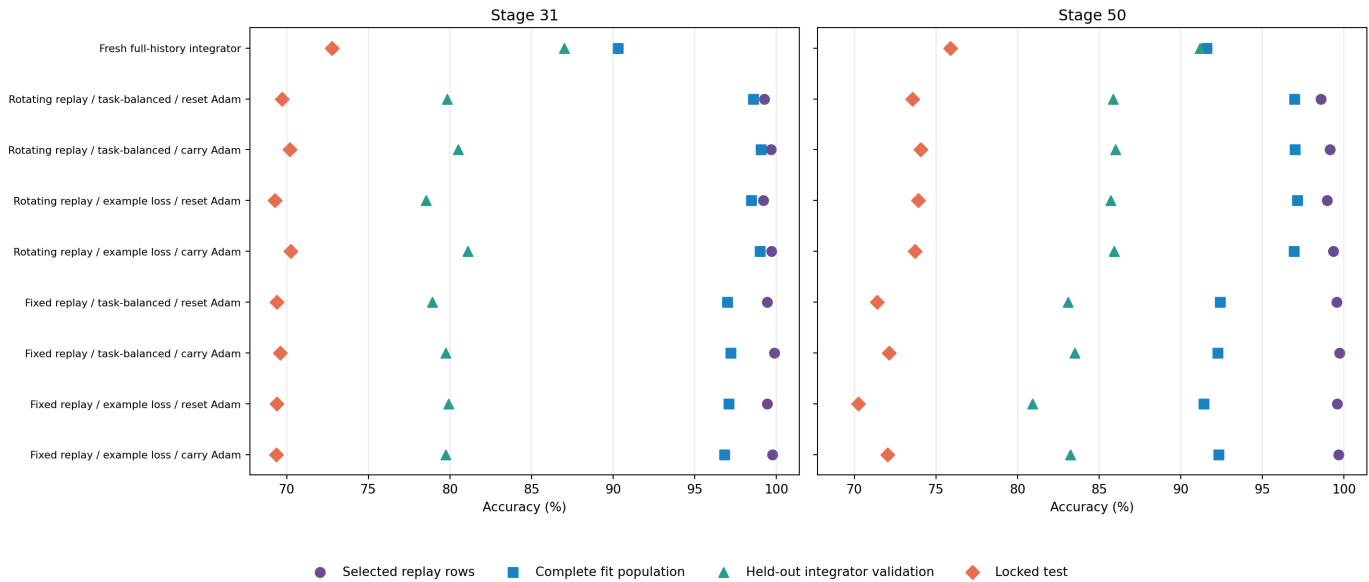
Novel identities and overlap relative to the immediately preceding historical subset.

Memorization and old-task adaptation

Stage	Condition	Selected train	Full fit	Validation	Test	Old-task test	Current-task test
31	Fresh full-history integrator	90.299	90.299	86.979	72.773	72.319	71.053
31	Rotating replay / example loss / carry Adam	99.685	98.991	81.076	70.231	69.607	76.316
31	Rotating replay / example loss / reset Adam	99.194	98.466	78.518	69.261	68.518	74.561
31	Rotating replay / task-balanced / carry Adam	99.673	99.057	80.485	70.178	69.636	73.684
31	Rotating replay / task-balanced / reset Adam	99.241	98.565	79.829	69.706	69.278	69.298
31	Fixed replay / example loss / carry Adam	99.755	96.818	79.731	69.340	68.668	73.684
31	Fixed replay / example loss / reset Adam	99.439	97.089	79.895	69.392	68.711	71.930
31	Fixed replay / task-balanced / carry Adam	99.871	97.195	79.731	69.575	68.841	74.561
31	Fixed replay / task-balanced / reset Adam	99.416	96.982	78.911	69.392	68.661	75.439
50	Fresh full-history integrator	91.573	91.573	91.167	75.867	75.300	78.182
50	Rotating replay / example loss / carry Adam	99.345	96.927	85.917	73.700	72.996	86.364
50	Rotating replay / example loss / reset Adam	98.947	97.146	85.688	73.917	73.209	83.636
50	Rotating replay / task-balanced / carry Adam	99.123	96.990	85.979	74.050	73.527	80.909

Stage	Condition	Selected train	Full fit	Validation	Test	Old-task test	Current-task test
50	Rotating replay / task-balanced / reset Adam	98.573	96.969	85.854	73.550	72.934	84.545
50	Fixed replay / example loss / carry Adam	99.661	92.323	83.229	72.017	71.202	86.364
50	Fixed replay / example loss / reset Adam	99.579	91.406	80.896	70.233	69.276	88.182
50	Fixed replay / task-balanced / carry Adam	99.719	92.250	83.479	72.117	71.353	82.727
50	Fixed replay / task-balanced / reset Adam	99.555	92.411	83.083	71.367	70.606	85.455

Memorization check: selected rows versus unseen populations

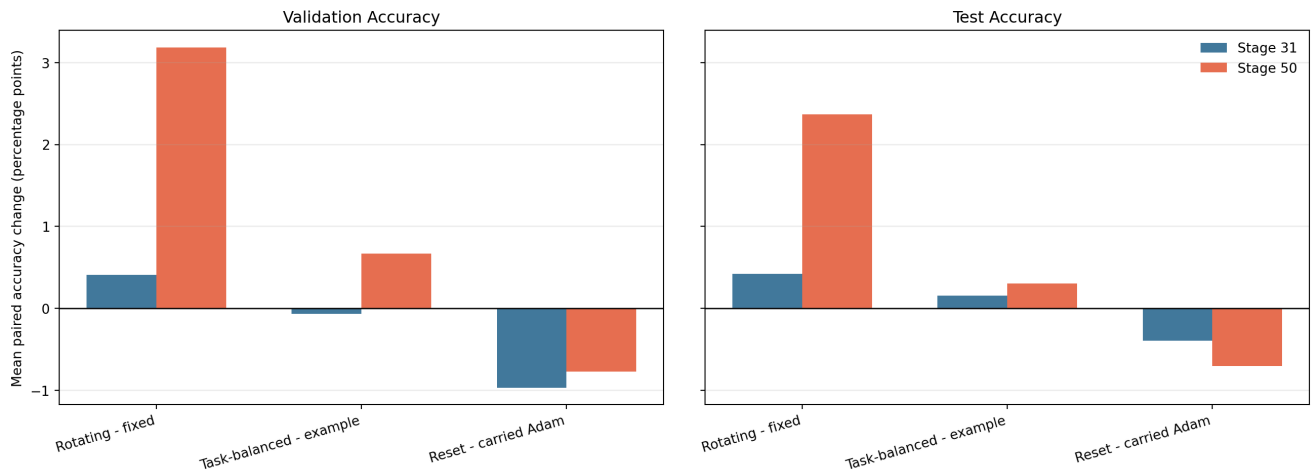


Selected replay rows can be memorized even when complete-fit, validation, and test accuracy remain much lower.

Paired factor effects

Factor	Metric	Mean paired change (pp)
reset_minus_carried_adam	test_accuracy	-0.549
reset_minus_carried_adam	validation_accuracy	-0.869
rotating_minus_fixed	test_accuracy	1.395
rotating_minus_fixed	validation_accuracy	1.799
task_balanced_minus_example	test_accuracy	0.231
task_balanced_minus_example	validation_accuracy	0.301

Main effects averaged over the other two paired factors



Each bar averages four exact paired contrasts over the other two factors.

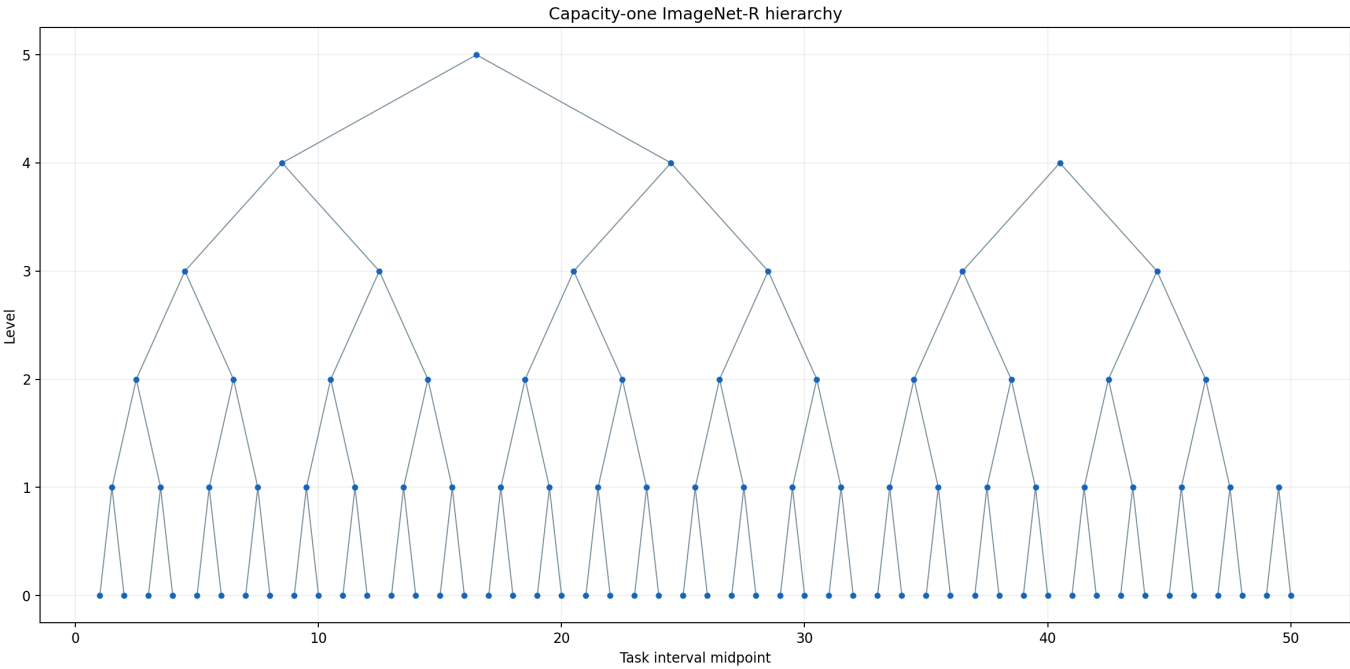
Interpretation boundaries

The fresh full-history fits use every fit row, three restarts, and validation-loss selection. They are diagnostic ceilings, not online LogT conditions. The integrator validation identities never enter integrator fitting, although the frozen LoRA nodes were previously fitted on all 24,000 training images. This is one seed and a previously examined test split, so test results are descriptive.

Reproducibility and work

Condition	Image presentations	Optimizer steps	Logical node/example bound	Fit seconds
Fixed replay / example loss / carry Adam	1375876	2760	1005091	14.990
Fixed replay / example loss / reset Adam	1375876	2760	1005091	14.310
Fixed replay / task-balanced / carry Adam	1375876	2760	1005091	14.396
Fixed replay / task-balanced / reset Adam	1375876	2760	1005091	14.418
Rotating replay / example loss / carry Adam	1375876	2760	1005091	14.403
Rotating replay / example loss / reset Adam	1375876	2760	1005091	14.434
Rotating replay / task-balanced / carry Adam	1375876	2760	1005091	15.447
Rotating replay / task-balanced / reset Adam	1375876	2760	1005091	15.310

Condition	Image presentations	Optimizer steps	Logical node/example bound	Fit seconds
Fresh full-history integrator	1883640	3720	-	49.123



The reused 50-task fresh-parent hierarchy ends with three live nodes; stages 31 and 50 expose five and three live nodes respectively.

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